ARPIT MISHRA

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SUMMARY

Artificial Intelligence Engineer with hands-on experience building LLM-powered backend systems, Retrieval-Augmented Generation (RAG) pipelines, agentic AI workflows, and knowledge graph applications using Python, FastAPI, and Neo4j.

EDUCATION

Bangalore Technological Institute (affiliated to VTU) *2022 - 2026*

B.E. Computer Science and Engineering with Artificial Intelligence and Machine Learning (AI/ML)

EXPERIENCE

AI Backend Engineer | National Informatics Centre (NIC) *Jul 2025 - Aug 2025 (on-site)*

- Designed and built a backend pipeline for an LLM-powered chatbot at NIC India, transforming 100% offline government PDF data into 5,000+ Subject-Predicate-Object triples ingested into a Neo4j knowledge graph.
- Developed an intent-classification and dynamic Cypher-generation module that improved query accuracy by 40%, cut response latency by 30%, and raised query success to 90%+, automating knowledge retrieval with zero manual intervention.
- Integrated backend services using FastAPI, LLM inference, RAG components, and Neo4j connectivity, with supporting React.js UI adjustments to expose new backend capabilities.

AI Backend Engineer, Remote Extension | National Informatics Centre (NIC) *Sep 2025 - Dec 2025*

- Extended the on-site engagement into a self-directed remote build, independently processing 40+ PDF documents end-to-end and scaling the knowledge-graph ingestion pipeline beyond the original scope.

Web Development Intern | Edutainer (PAT Technologies Pvt. Ltd.) *Feb 2026 - May 2026*

- Built and deployed a full-stack business/portfolio-style website using HTML, CSS, and JavaScript on the frontend with a Node.js backend.
- Implemented responsive UI components and core backend routes/logic to support site functionality, gaining hands-on experience with the end-to-end web development lifecycle.

PROJECTS

Enterprise Hybrid Graph-RAG & Stateful Agent System | GitHub *Aug 2026 - Sept 2026*

LangGraph, FastAPI, Neo4j, Redis, Docker, Streamlit, OpenAI GPT-4o-mini, Ollama

- Engineered a self-correcting Text-to-Cypher retrieval pipeline achieving a 100.0% exact match rate across 36 benchmark suites spanning aggregation, multi-hop traversal, and schema edge-case queries, by building a validator node that auto-detects syntax/directionality errors and triggers regeneration before execution.
- Architected a dynamic few-shot prompting system eliminating hallucinated relationship paths across a 7-node, 6-relationship graph ontology (10,000+ visits, 9,650 patients), via a Neo4j vector store (k=8) retrieving relevant verified Cypher examples at inference time.
- Mitigated database security and stability risks, achieving a 100% block rate against mutating queries (CREATE, DELETE, MERGE, DROP) and zero DOS incidents, by enforcing pre-execution AST/regex guards, read-only access isolation, and mandatory query result ceilings.
- Optimized query response latency for repeat requests, reducing average latency to ~0.10 ms, by implementing an SHA-hashed, normalization-aware Redis L1 caching layer that bypasses LLM inference entirely.
- Benchmarked end-to-end system reliability, achieving ~10.14 ms average query execution time and a 100% execution success rate with 0 runtime graph errors, by designing a deterministic 4-pathway LangGraph state machine with SSE-based sub-second token streaming.

AI-Powered Excel Report Generator | GitHub *Oct 2025 - Feb 2026*

Python, FastAPI, LLM Integration (Gemini), React.js

- Engineered a full-stack application transforming natural-language input into structured Excel reports via prompt engineering and LLM-based data extraction, cutting manual report creation time by ~70%.
- Designed a prompt-to-schema pipeline supporting 10+ input fields per report with 90%+ field-mapping accuracy, built on a React.js frontend and FastAPI backend.

AI Hover Mouse | GitHub *Sep 2024 - Dec 2024*

Python, OpenCV, MediaPipe

- Developed a real-time hand-gesture-based mouse controller using computer vision, achieving 95% cursor accuracy and sub-50ms response latency for smooth on-screen interaction.
- Implemented optimized gesture recognition and noise-reduction algorithms, reducing false detections by 40% and improving tracking stability across varying lighting conditions.

TECHNICAL SKILLS

Backend Development: Python, FastAPI, RESTful APIs, LLM Integration, RAG Pipelines, Agentic AI Workflows, Neo4j, Cypher Query Language

AI / Machine Learning & NLP: Generative AI, Intent Classification, Entity Relation Extraction, Knowledge Graph Modeling, SPO Triple Generation, Embeddings, Vector Databases, Prompt Engineering, PyTorch

Frontend: React.js, JavaScript, HTML, CSS, Bootstrap

Deployment Environment: Linux, Server Deployment (NIC Internal), API Testing, Debugging & Optimization

Databases & Tools: MongoDB, MySQL, SQL, Docker, Kubernetes, Postman, Vercel, Neo4j Browser, PyPDF / PDFMiner, Git, GitHub

Other Skills: Data Cleaning, Unstructured Data Processing, Knowledge Graph Ingestion, System Integration

CERTIFICATIONS

- The Ultimate Job Ready Data Science Course - CodeWithHarry
- Web Development Internship Completion - Edutainer (PAT Technologies Pvt. Ltd.)

PUBLICATIONS

- "NLP Powered Excel Assistant: A Smart Chatbot for Automated Data Processing" - IRJMETS, Vol. 7, Issue 9, September 2025

ACHIEVEMENTS & RESPONSIBILITIES

- Workshop Host, "Intro to Deep Learning": Led a hands-on session for 80+ students on neural networks, backpropagation, and PyTorch.